

This guide is intended for those who have yet to enter university or take part in physics as it is typically presented at an institution.

I love mathematics, and I think one should not overlook it in favor of physics. This PDF covers the mathematics you will typically complete before your freshman year of college, along with some extras here and there for the mathematically inclined. You may find many topics are repeated. This is deliberate so that you can get exposed to different styles: applied and theoretical.

If you have a better or equivalent alternative, use that. You may also download a PDF where available, or obtain the books by any other means. There is no need to waste money.

1. We begin with basic arithmetic and prepare for a course in algebra. This book is quite good and non-routine. Use *Prealgebra* by Rusczyk.
2. The Gelfand Correspondance program, the For the Practical Man series as well as other books for a first pass in basic mathematics.
 - *Algebra* by Gelfand and Shen and *Algebra for the Practical Man*
 - *The Method of Coordinates* by Gelfand, Glagoleva, and Kirillov
 - *Functions and Graphs* by Gelfand, Glagoleva, and Shnol
 - *Trigonometry* by Gelfand and Saul and *Trigonometry for the Practical Man*
 - *Elements of Coordinate Geometry* by S.L. Loney
 - *Geometry* by Gelfand and Alekseyevskaya and *Geometry for the Practical Man*
3. Next we learn how to do basic proofs in mathematics.
 - *How to Prove It* by Velleman
 - *Book of Proof* by Hammack
4. Having done basic proofs we can now look into discrete mathematics.
 - *Discrete Mathematics* by Epp
 - *Introductory Discrete Mathematics* by Balakrishnan
 - *Combinatorics* by Mladenovic
 - *Principles and Techniques in Combinatorics* by Chen, Koh
5. Alongside discrete mathematics you may do these books too.
 - *Introduction to Linear Algebra* by Lang
 - *About Vectors* by Hoffman
 - *Elementary Number Theory* by Burton
 - *Complex Numbers* by Andresscu, Andrica
6. We review all mathematics we have learnt as well as start calculus.
 - *Basic Mathematics* by Lang
 - *Precalculus* by Axler
7. Well we now do calculus and introductory analysis. While high schoolers or incoming freshman don't really do this much I think that if you have time you may as well do these books in order I present them. *Calculus: A Complete Course* by Adams, Essex and *Elementary Differential Equations and Boundary Value Problems* by Boyce, DiPrima is enough for an applied calculus course for students who want to study applied physics or do not want to learn this much math. The mathematics from top to here is enough for introductory physics. The following texts are too advanced but nonetheless very interesting and doing the first one (Spivak) is enough to check whether you like or dislike mathematics. If you do not want to do advanced mathematics like analysis or modern algebra and need it only to pursue physics then refer *Mathematical Methods in the Physical Science* by Boas for all your undergraduate needs.

- *Calculus: A Complete Course* by Adams, Essex
- *Elementary Differential Equations and Boundary Value Problems* by Boyce, DiPrima
- *Calculus* by Spivak
- *Vector Calculus, Linear Algebra, and Differential Forms* by Hubbard, Hubbard
- *Calculus on Manifolds* by Spivak and *Calculus on Manifolds* by Munkres
- *How to Think About Analysis* by Alcock
- *Understanding Analysis* by Abbott

Well a guide is incomplete without recommending problem books for the various topics mentioned here.

1. *Problems in Elementary Mathematics* by Lidsky
2. *A Problem Book in Algebra* by Krechmar
3. *104 Trigonometry Problems* by Adreescu, Andrica
4. *Problems in Plane Geometry* by Sharygin and *Problems in Plane Geometry* by Prasolov
5. *104 Number Theory Problems* by Andreescu, Andrica, Feng and *Problems in Number Theory* by Sierpinski
6. *Problems in Complex Numbers* by Prasolov
7. *The USSR Olympiad Problem Book* by Shklyarsky, Chentkov, Yaglov
8. *Problem-Solving through Problems* by Larson
9. *Problems and Theorem in Linear Algebra* by Prasolov
10. *Problems in Algebra, Arithmetic and Analysis* by Prasolov
11. *Problems in Calculus of One Variable* by I.A. Maron
12. *Calculus with Multiple Variables* by McMullen
13. *Problems in Analytic Geometry* by Kletenik
14. *Problems in Mathematics* by Govorov, Dybov, Miroshin, Smibnova
15. *A Problem Book in Mathematical Analysis* by Berman